

William Heath

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Summary of Qualifications

- Freshman in Computer Science at Cal Poly San Luis Obispo.
- Diverse programming experience from Web Dev to Game Dev, to robotics

Education

California Polytechnic State University, San Luis Obispo | B.S. in Computer Science | GPA: 3.90 **Expected June 2029**

- **Scheduled Coursework:** Data Structures, Project-Based OOP & Design, Design & Analysis of Algorithms

Garfield High School, Seattle, WA | Valedictorian | GPA: 4.00

Sep 2020 – June 2025

Skills

Languages: Java, C#, Python, JavaScript, TypeScript, HTML, CSS

Frameworks/Other: React, Tailwind CSS, Unity, Git

Skills: Robotics, Web development, Game Dev, Data Structures & Algorithms, Design Patterns

Experience

Microsoft Hunt The Wumpus Competition Winner

Feb 2025 – May 2025

- Designed a dungeon-crawler game from scratch in Java, playable in the Command-Line Interface
- Followed project criteria and applied knowledge of design patterns, focusing on scalable architecture and minimizing internal dependencies; abstracted systems like dungeon generation using the strategy pattern to enable quick swapping for prototyping
- Implemented custom ASCII rendering engine with Z-indexing

Garfield Robotics Autonomous Lead – FTC Team 4042

Sep 2023 – June 2025

- Built an autonomous movement system focusing on reusable façade components for future team members
- Developed System for queueing action groups, enabling components such as the drive base, lift, and claw to function in tandem
- Drive base movement leveraged a PID control loop built and tuned from scratch for accurate and fast transportation
- Gained communication and teamwork skills while developing alongside a mechanical team to meet deadlines

Game Development

Sep 2020 – Present

- Designed a multiplayer 2D Arena Fighter using TCP/UDP servers for ongoing matches and a Load Balancer that directs players to horizontally-scalable match manager servers using a REST API
- Implemented Singleton, Observer, Flyweight, Factory, and Strategy design patterns often when creating performant and maintainable code.
- Developed primarily in Unity, with some experience in Unreal Engine 4 and Game Maker Studio 2

Employment

Professional Jazz Pianist

Sep 2020 – June 2025

- Played 50+ gigs; hired for corporate events, weddings, bar mitzvah, and private parties

Mad Candy

Aug 2023 – Mar 2025

- Customer sales at local candy store

Awards & Honors

- Eagle Scout – Troop 167
- National Jazz Competition Essentially Ellington Honorable Mention Rhythm Section
- Microsoft Hunt The Wumpus Competition Best Implementation and Crowd Favorite

Hobbies

Pianist

- I have worked with a variety of groups and genres, playing piano for my high school's jazz band, choir, and theatre, as well as playing professionally

Outdoor Activities

- Skiing, hiking, mountain biking, windsurfing